Blockchain & Secure Multi-Party Computation

Master 2 — Computer Science

Session 1 of 12 — Introduction to Blockchain I

Course Roadmap

This course builds expertise in two transformative technologies: blockchain and secure multi-party computation (MPC). The first half (Sessions 1–6) covers blockchain from its fundamental principles through its architecture, programming model, security, and interoperability. The second half (Sessions 7–10) covers smart contracts and dApps. Sessions 11–12 introduce MPC, its theoretical foundations, and its powerful combination with blockchain. The table below gives the full overview.

Session	Topic
1–2	Introduction to Blockchain: principles, history, types, applications
3–4	Cryptography for Blockchain: hash functions, signatures, encryption, consensus
5–6	Blockchain Structure: architecture, transaction validation, scalability
7–8	Smart Contracts and Decentralized Applications
9	Blockchain Security and Privacy
10	Interoperability and Cross-Chain Blockchain
11–12	Secure Multi-Party Computation: foundations, IBE, ABE, protocols

1 What Is a Blockchain?

Definition — Blockchain

A blockchain is a continuously growing list of records, called blocks, that are linked together and cryptographically secured to form a chain. Each block contains: a cryptographic hash of the previous block (its "fingerprint"), a timestamp, and a set of transaction data. The chain is stored simultaneously across a peer-to-peer network of computers (nodes), with no single node acting as a central authority. Formally:

$$B_i = \big(H(B_{i-1}),\ t_i,\ \text{data}_i,\ \text{nonce}_i\big), \qquad H(B_i) = \text{SHA256}(\text{SHA256}(B_i))$$

where H is a cryptographic hash function, t_i is the timestamp, and the nonce is a value used in mining (covered in Session 3).

The key insight is that the hash of each block is embedded in the next one. This means that if someone alters a past block, its hash changes, which breaks its link to the next block, which in turn cascades through every subsequent block. The entire network would immediately reject this altered chain because it would no longer match the majority's version.

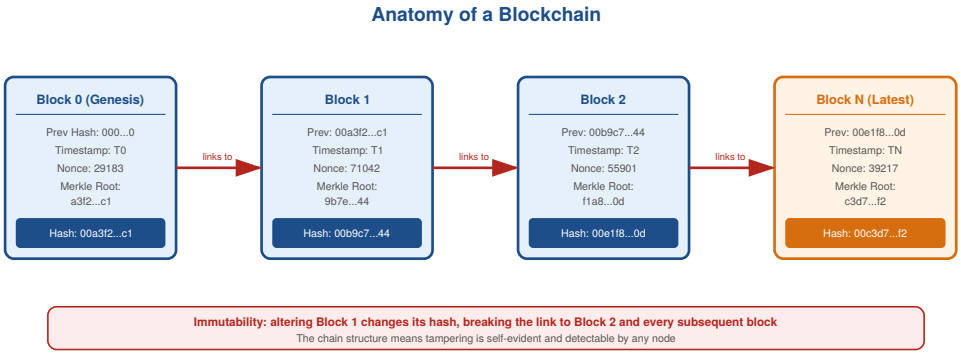


Figure 1: The structure of a blockchain: each block contains the hash of the previous block, so altering any block breaks all subsequent links and is immediately detectable.

? Think about it

Why does embedding the hash of the previous block make the chain tamper-resistant?

✓ Answer

Because hash functions are one-way and collision-resistant (see Session 3): changing even one byte of a block completely changes its hash. This modified hash no longer matches what the next block stores as "Prev Hash," breaking the chain. An attacker would need to recompute every subsequent block's hash faster than the entire honest network is adding new blocks — computationally infeasible in a large network.

2 Fundamental Principles of Blockchain

Blockchain technology rests on five foundational principles that together create a system where trust between parties can be established without any central authority.

2.1 Decentralization

Definition — Decentralization

In a decentralized system, no single entity controls the data or the rules. Every participant (node) holds a complete copy of the blockchain and participates in validating new transactions. Control is distributed among all nodes.

Example — Decentralization in Bitcoin

Bitcoin's network consists of tens of thousands of nodes worldwide. No government, company, or individual owns or controls Bitcoin. If any single country shuts down its local Bitcoin nodes, the network continues operating globally without interruption. Compare this to a bank, whose database is centrally owned and can be frozen, confiscated, or censored by authorities.

? Think about it

Give an example of the benefit of decentralization.

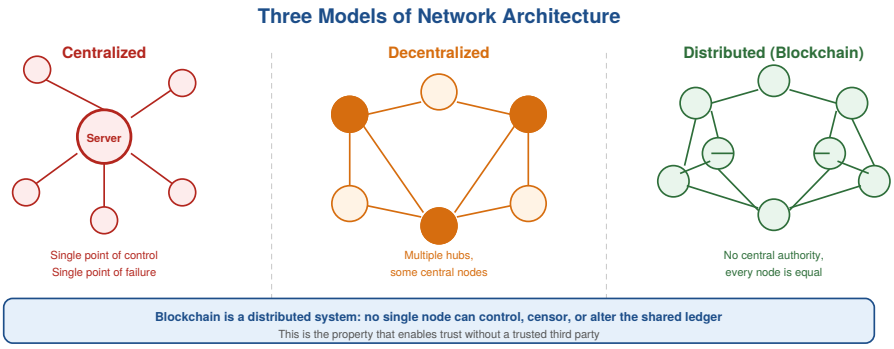


Figure 2: Three network models. Classical systems are centralized (left): one server controls everything. Blockchain uses the distributed model (right): every node is equal, holds the full ledger, and participates in validation.

✓ Answer

In 2021, China banned all cryptocurrency mining and trading. Bitcoin's mining activity temporarily dropped, but within months, miners had relocated to other countries and the network continued functioning normally. No shutdown command exists because there is no central server to shut down.

2.2 Immutability

Definition — Immutability

Once a block is added to the chain and confirmed by subsequent blocks, the data it contains cannot practically be altered. The cryptographic hash chain makes retroactive modification computationally infeasible.

Example — Immutability and audit trails

A supply chain company uses a blockchain to record each step in a product's journey from factory to consumer. Because the records are immutable, no participant can secretly alter a date, a location, or a quality certification after the fact. Every record is permanently and publicly auditable.

? Think about it

Is blockchain truly 100% immutable? Can anything be changed after the fact?

✓ Answer

Not with zero cost. Changing a past record requires recomputing all subsequent blocks' proofs-of-work and controlling more than 50% of the network's computing power (a "51% attack") — extremely expensive for large networks like Bitcoin. However, small blockchains with few nodes are more vulnerable. There are also rare cases where a community votes to fork the chain to reverse a transaction (as happened with Ethereum after the 2016 DAO hack), showing immutability is a practical rather than absolute guarantee.

2.3 Transparency

In most public blockchains, every transaction ever recorded is visible to anyone. Any participant can download the full blockchain and verify its entire history independently, without trusting any

other party. This transparency eliminates the need for auditors or trusted intermediaries.

Example — Transparency in practice

Any user can go to `blockchain.com` or `etherscan.io` and look up any Bitcoin or Ethereum transaction ever made — who sent funds, how much, when, and to which address. No bank provides this level of public verifiability for its transactions.

? Think about it

If every transaction is visible, does blockchain protect user privacy?

✓ Answer

Blockchain offers pseudonymity, not anonymity. Addresses (public keys) are visible, but they are not directly linked to real-world identities. However, through blockchain analysis (tracing transaction flows), it is often possible to de-anonymize users, especially when they interact with exchanges that require identity verification. Privacy-focused blockchains like Monero or Zcash add cryptographic techniques to hide amounts and addresses.

2.4 Security Through Cryptography

Every aspect of blockchain operation relies on cryptographic primitives covered in Sessions 3–4: hash functions protect block integrity, digital signatures authenticate transactions, and consensus mechanisms (Proof of Work, Proof of Stake, etc.) ensure agreement among nodes.

Example — Why no password is needed to "own" Bitcoin

Bitcoin ownership is controlled not by a username/password on a server, but by a private key. To spend Bitcoin from an address, the owner must produce a valid digital signature using their private key. No central server stores or checks this — any node in the network can independently verify the signature's validity.

2.5 Consensus

Definition — Consensus Mechanism

A consensus mechanism is a protocol by which all nodes in a distributed network agree on a single version of the blockchain, even in the presence of faulty or malicious participants. It answers the question: "which block gets added next, and who decides?"

The consensus mechanism is what replaces the central authority. Instead of a bank deciding whether a transaction is valid, the consensus mechanism ensures that the network as a whole reaches agreement, automatically and without trust in any single participant. The main mechanisms (PoW, PoS, BFT, Proof of Chance) are studied in depth in Session 4.

? Think about it

Why is consensus the hardest problem in blockchain design?

✓ Answer

Because participants may be geographically distributed, may have different versions of the current state, and some may actively try to cheat. The Byzantine Generals Problem (also covered in Session 4) formalizes this: how can a group reach agreement when some members may be traitors sending conflicting messages? Blockchain consensus mechanisms are practical solutions to this classical problem.

3 History and Evolution of Blockchain Technology

Understanding blockchain's history reveals why each version was created and what problem it was designed to solve.

3.1 Pre-Bitcoin Foundations (1970s–2000s)

Blockchain's building blocks were developed long before Bitcoin:

- **1976**: Diffie and Hellman propose public-key cryptography.
- **1991**: Stuart Haber and W. Scott Stornetta propose timestamping documents using a cryptographic hash chain — the direct conceptual precursor to blockchain.
- **1998**: Nick Szabo proposes "Bit Gold," a decentralized digital currency concept using proof-of-work.
- **2001**: Adam Back formalizes Hashcash, a proof-of-work system to combat email spam — later adopted by Bitcoin.

3.2 Blockchain 1.0: Bitcoin (2008–2014)

Example — The Bitcoin white paper

In October 2008, during the global financial crisis, an anonymous person (or group) using the pseudonym Satoshi Nakamoto published "Bitcoin: A Peer-to-Peer Electronic Cash System." In January 2009, the first Bitcoin block (the "Genesis Block") was mined, containing the message: "The Times 03/Jan/2009 Chancellor on brink of second bailout for banks" — a direct commentary on the problem Bitcoin was designed to solve.

Bitcoin demonstrated that a decentralized digital currency was possible. This era is called "Blockchain 1.0": blockchain as a ledger for cryptocurrency transactions.

3.3 Blockchain 2.0: Ethereum and Smart Contracts (2015–2018)

Vitalik Buterin, a teenager reading about Bitcoin, asked: why can the blockchain only record currency transfers? What if it could run arbitrary programs? In 2015, Ethereum launched, introducing the concept of smart contracts — self-executing code stored on the blockchain. This unlocked an entirely new category of applications.

3.4 Blockchain 3.0: Scalability, Interoperability, and Beyond (2018–present)

The third generation addresses limitations of earlier blockchains: slow transaction speed, high energy consumption, inability of different blockchains to communicate, and the need for real-world data (handled by "oracle" services). Key projects of this era include Polkadot, Cosmos, Cardano, and Solana.

Era	Period	Representative	Key innovation
1.0	2008–2014	Bitcoin	Decentralized digital currency
2.0	2015–2018	Ethereum	Smart contracts & programmable blockchain
3.0	2018–now	Polkadot, Solana	Scalability, interoperability, sustainability

? Think about it

Why is blockchain development moving so fast? What drives this evolution?

✓ Answer

Two forces. First, each version exposes practical limitations (Bitcoin is too slow for general use; Ethereum is too expensive when congested), creating demand for improvement. Second, because most blockchain projects are open-source and their value is publicly measurable in real time (token prices, usage statistics), the feedback loop between developers and markets is much faster than in traditional software development.

4 The Double-Spend Problem: Why Blockchain Was Necessary

The fundamental problem that motivated blockchain's invention is worth understanding precisely, because it explains why every other design decision follows.

Definition — The Double-Spend Problem

In digital systems, a file can be copied perfectly. If money is a digital file, what prevents someone from copying it and spending it twice? In traditional banking, a central server maintains an authoritative ledger and prevents double-spending. In a decentralized system with no trusted server, this problem was unsolved until Bitcoin.

Example — How Bitcoin solves double-spending

Alice has 1 Bitcoin. She simultaneously sends it to Bob and to Carol (attempting double-spend). Both transactions propagate through the network. However, only one can be included in a block — miners select transactions and include them in the blockchain. The transaction that gets included first becomes the valid one; the other is rejected. Because the entire network sees and agrees on the blockchain, there is no way to successfully spend the same coin twice once a transaction is confirmed.

? Think about it

Does blockchain eliminate double-spending completely?

✓ Answer

Almost. A transaction is considered "final" after a certain number of confirmations (subsequent blocks added after it). With 6 confirmations in Bitcoin, a reversal requires an attacker to outpace the entire network's hash power for that time window — costing hundreds of millions of dollars on the Bitcoin network. For practical purposes this is secure, but technically it is probabilistic rather than absolute.

5 Who Participates: Nodes and Their Roles

A blockchain network is composed of multiple types of participants, each playing a distinct role. Understanding these roles clarifies how the network functions without any central administrator.

Definition — Full Node

A full node downloads, stores, and independently validates every block and every transaction in the blockchain's entire history. It enforces the consensus rules and does not trust any other participant's claims — it verifies everything itself. Full nodes are the backbone of the

network's security.

Definition — Miner / Validator Node

In Proof of Work blockchains, a miner is a node that competes to produce the next valid block by solving a computational puzzle (covered in Session 4). In Proof of Stake blockchains, validators are nodes chosen proportionally to their stake (the amount of cryptocurrency they have locked up) to propose and attest to new blocks. Both roles earn rewards for honest participation.

Example — Lightweight (SPV:Simplified Payment Verification) nodes

Most smartphone wallets are lightweight nodes. They do not download the full blockchain; instead, they download only block headers and use Merkle proofs (Section 4 of this session) to verify that specific transactions are included in confirmed blocks. This reduces storage from hundreds of gigabytes to a few megabytes at the cost of trusting that full nodes have validated the full transaction history.

? Think about it

If a miner produces a block with an invalid transaction (e.g., spending coins that do not exist), what happens?

✓ Answer

Other full nodes, upon receiving this block, independently validate every transaction in it. The invalid transaction will fail validation — it violates the consensus rules. Every honest full node will reject this block. The miner who produced it will have wasted their computational effort and receive no reward. This is the self-enforcement mechanism: dishonest mining is expensive and produces no benefit.

6 The Merkle Tree: Efficiently Summarizing All Transactions

Each block does not simply list all transactions linearly; it organizes them in a Merkle tree structure, which allows any node to efficiently verify that a specific transaction is included in a block without downloading all transactions.

Definition — Merkle Tree

A Merkle tree is a binary hash tree where: leaf nodes contain the hash of individual transactions ($h_i = H(\text{tx}_i)$); each internal node contains the hash of its two children ($h_{AB} = H(h_A \| h_B)$); and the root node (the Merkle Root) is a single hash summarizing all transactions. If any transaction changes, the root changes completely.

$$\text{MerkleRoot} = H(H(H(\text{tx}_1)\|H(\text{tx}_2))\|H(H(\text{tx}_3)\|H(\text{tx}_4)))$$

Example — Why the Merkle tree matters for lightweight clients

A smartphone cannot store the entire Bitcoin blockchain (500+ GB). Using the Merkle tree, a lightweight client (SPV — Simplified Payment Verification) can verify that a specific transaction is included in a block by downloading only the transaction itself plus a "Merkle proof" (the sibling hashes along the path to the root) — around 10–12 hashes instead of hundreds of thousands of transactions.

? Think about it

If a hacker could modify one transaction in the middle of a Merkle tree, how many hash values would need to be recomputed to hide the tampering?

✓ Answer

All hashes along the path from that leaf to the root — roughly $\log_2(n)$ values where n is the number of transactions. But the Merkle Root is stored in the block header, and the block header's hash is embedded in the next block. So successfully hiding the change also requires changing the block header, which changes the block's hash, which breaks the chain — the tampering cascades upward and becomes visible to all nodes.

7 Wallets and Addresses: How Users Interact

Users do not interact with a blockchain by logging in with a username and password. Instead, they use a cryptographic key pair.

Definition — Blockchain Wallet

A blockchain wallet does not "store" coins the way a physical wallet stores cash. Instead, it stores the user's private key, which proves ownership of an address. The coins (or tokens) themselves exist as entries in the blockchain ledger. A wallet manages the private key and creates signed transactions.

Example — A Bitcoin address as a public identity

A Bitcoin address (e.g., `1A1zP1eP5QGefi2DMPTfTL5SLmv7Divf Na`) is derived from a public key by applying SHA-256 followed by RIPEMD-160 and a checksum. It is the "account number" that anyone can send funds to. Only the holder of the corresponding private key can sign a transaction spending those funds. If the private key is lost, the funds are permanently inaccessible — no bank or administrator can reset it.

? Think about it

A user loses the piece of paper where they wrote their Bitcoin private key, and their computer with the wallet software crashes. What happens to their Bitcoin?

✓ Answer

The Bitcoin remains at that address on the blockchain — permanently. Without the private key, no valid signature can be produced, and no transaction can be authorized. The coins are effectively frozen forever. This is why secure key management (hardware wallets, seed phrase backup) is critically important. Approximately 20% of all Bitcoin is estimated to be permanently lost this way.

Example — Seed phrases: backing up a key without writing raw binary

Modern wallets use a "seed phrase" (e.g., 12 or 24 common English words like "witch collapse practice feed shame open despair creek road again ice eager") that encodes the private key in human-readable form. This phrase can be written on paper and stored securely, and any compatible wallet software can restore the full wallet from it.

8 Key Properties: A Summary Comparison

It is useful to compare blockchain against traditional database systems across the properties that matter most:

Property	Traditional Database	Blockchain
Control	Centralized owner	Distributed among all nodes
Modification	Administrator can modify any record	Cryptographic hash chain makes past records practically immutable
Trust model	Trust the database owner	Trust the protocol and mathematics
Transparency	Owner decides what is visible	All transactions publicly verifiable (public chains)
Fault tolerance	Single point of failure	Survives loss of many nodes
Consensus	ACID transactions decided by one system	Network-wide consensus protocol

? Think about it

Given these trade-offs, is blockchain always better than a traditional database?

✓ Answer

No. Blockchain is slower, more expensive to operate, and harder to modify than a traditional database. If you trust a central authority (e.g., a private internal company database where employees have no reason to deceive each other), a traditional database is faster, simpler, and more efficient. Blockchain adds value specifically when multiple mutually distrustful parties need to share data without trusting each other or a common intermediary.

Session Summary

This session established the foundational vocabulary of blockchain: what a blockchain is (a cryptographically linked chain of blocks stored on a distributed network), its five fundamental principles (decentralization, immutability, transparency, cryptographic security, and consensus), the history of its development from Bitcoin through Ethereum to the current generation, and the specific problem (double-spending) that motivated its invention. The Merkle tree structure shows how transaction data is efficiently organized inside each block, and the comparison table clarifies that blockchain's value is specific: it replaces trust in a central authority with trust in mathematics and distributed consensus — a trade-off that makes sense when participants cannot or should not trust any single party to operate a database on their behalf. Session 2 covers blockchain types (public, private, consortium, hybrid) and the wide range of sectors where these properties create real value.

? Think about it

In one sentence, what is the core social promise of blockchain technology?

✓ **Answer**

Blockchain allows parties who do not trust each other — and have no trusted intermediary they both accept — to nonetheless agree on a shared, tamper-evident history of events, through mathematics and distributed consensus rather than through authority or reputation.